ORIGINAL ARTICLE

Longevity and Complication Rates of Ultrasound Guided Versus Traditional Peripheral Intravenous Catheters in a Pediatric Emergency Department

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Abstract

Background: *Ultrasound-guided peripheral intravenous lines are frequently used in patients with difficult access. We have previously reported on the longevity and complication rates of ultrasound-guided peripheral intravenous lines, but there are limited data comparing outcomes of ultrasound-guided peripheral intravenous lines to traditionally placed peripheral intravenous lines in children. The aim of this study was to compare the longevity and complication rates of ultrasound-guided peripheral intravenous lines to traditionally placed intravenous lines in a pediatric population.*
Methods: *This study analyzed 300 ultrasound-guided peripheral intravenous lines and 552 traditionally placed intravenous lines using patient records to determine the reason and timing for intravenous line removal. A* t*-test was used to compare overall mean survival times, and a log-rank test was used to compare Kaplan-Meier survival curves. Complication rates were compared using a chi-squared test.*
Results: *The survival times of ultrasound-guided peripheral intravenous lines (mean = 73 hours, SD = 68 hours) were significantly longer than those of traditionally placed intravenous lines (mean = 38 hours, SD = 29.4 hours),* $t(559) = 8.51$, $P < .0001$. *Kaplan-Meier survival analysis yielded a median ultrasound-guided peripheral intravenous line survival time of 143 hours (IQR = 68-246) and a median traditionally placed intravenous line survival time of 100 hours (IQR = 65-106) with a significant difference between the 2 survival curves by the log-rank test. There was also no significant difference in complication rates between ultrasound-guided peripheral intravenous lines (34.8%) compared to traditionally placed intravenous lines (31.8%),* $x^2(1, N = 517) = 0.465$, $P = .50$.
Conclusions: *Our data suggests that ultrasound-guided peripheral intravenous lines are a viable option for children, including those with a history of difficult access. Survival times were longer for ultrasound-guided peripheral intravenous lines versus traditionally placed intravenous lines, and complication rates of the ultrasound-guided peripheral intravenous lines and traditionally placed intravenous lines were similar.*
Keywords: *ultrasound, intravenous access, pediatrics, emergency care*

Background

Problem Description

A current challenge in emergency departments (EDs) is obtaining peripheral intravenous (IV) access. Placing peripheral IVs with ultrasound guidance has emerged as a way to decrease the number of needlesticks. The placement of peripheral IVs in children is particularly difficult due to decreased patient compliance during placement and the smaller size of veins. Because multiple failed attempts at IV placement can be particularly distressing for children and their parents, a variety of techniques have been used to aid in the visualization of peripheral veins, including local warming of IV site and more advanced technologies such as transillumination, near-infrared lighting, and ultrasound.[1] Some emergency departments have instituted teams of nurses and doctors that are IV specialists to maximize IV insertion success rates.[2]

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Available Knowledge

To determine whether IVs placed with real-time ultrasound visualization leads to fewer failed IV attempts, previous clinical studies have compared success rates of traditionally placed peripheral IVs (TPIVs) to ultrasound-guided peripheral IVs (USGPIVs). In 2012, a randomized controlled trial (RCT) was conducted comparing USGPIVs to TPIVs in children under 3 years of age with difficult venous access. In the study, USGPIVs had a significantly lower median time to cannulation and median number of punctures in addition to a higher success rate at first cannulation.[3] A 2009 study performed a RCT of children under 10 years of age in a pediatric ED who required IV access and had either 2 previous unsuccessful traditional attempts at IV placement or a history of difficult access. The patients were randomized to undergo IV placement by either continued traditional attempts or with live ultrasound visualization. The USGPIVs required less overall time to place, fewer attempts at placement, and fewer needle redirections than TPIVs.[4]

We previously analyzed USGPIV attempts in a pediatric ED to determine the success rate, longevity, and complications of USGPIVs. The complications of USGPIVs were similar to those commonly reported for TPIVs, including unintentional dislodgement, infiltration, occlusion, and phlebitis. In terms of the longevity, the study found that the Kaplan-Meier median survival time of USGPIVs was 143 hours (6 days).[5] Previous studies in adults have shown poor durability of USGPIVs. Some attribute the poor durability to ultrasound assistance allowing cannulation of deeper and smaller vessels; however, there are limited data on the outcomes of USGPIVs directly compared to TPIVs.[6-8]

Rationale and Specific Aims

As demonstrated by previous studies, ultrasound guidance is both an efficient and effective aid for IV placement especially in patients with difficult access. Determining whether USGPIVs have higher complication rates and poorer durability compared to TPIVs will affect the feasibility of using ultrasound in pediatric EDs to place IVs. The specific aim of this study was to directly compare the longevity and complication rates of USGPIVs to TPIVs in a pediatric ED.

Methods

Context and Intervention

The study took place in an urban tertiary-care teaching hospital with a pediatric ED that sees approximately 97,000 patients per year. A high percentage of the patients have chronic illnesses and have issues with IV access. For patients that have had a history of difficult access or had failed IV attempts via the traditional approach, ultrasound is often used to facilitate IV placement.

From August 2013 to April 2014, a formalized training program for USGPIVs took place in the ED for nurses and physicians as part of a hospital-wide difficult access initiative. ED practitioners placing USGPIVs included nurses, pediatric and emergency medicine (EM) residents, pediatric emergency medicine (PEM) fellows, and PEM attendings. The EM residents, PEM fellows, and PEM attendings all completed a rotation in emergency bedside ultrasound before being authorized to place USGPIVs. The nurses and pediatric residents underwent a 4-hour course that had both a didactic component and hands-on training. Any USGPIVs placed during this 4-hour training period were also included in the data set.

Methods and Measures

This study analyzed data from a data set collected as part of an initial quality improvement project and was deemed exempt by the hospital's institutional review board. For each USGPIV placed from August 2013 to April 2014 as part of the training program detailed earlier, the practitioner placing the IV was asked to complete a form documenting why the IV was placed, if traditional IV access was attempted along with the number of TPIV attempts, and the number of USGPIV attempts along with whether or not the USGPIV was successfully placed. The reasons for USGPIV placement included history of difficult IV access, multiple failed TPIV attempts, patient or family request, and teaching purposes. The form included identifying patient information to track the USGPIVs in the electronic health record (EHR). All USGPIVs that were placed by an ED clinician participating in the training program were eligible for inclusion in the study. Two providers (A.E.C., A.M.V.) performed the follow-up in the EHR for all of the USGPIVs that were placed in the study. Removal reason, complications of the IV, and time from IV placement to removal was extracted from the EHR.

The USGPIVs were placed via the dynamic method in the short axis. The provider decided both the placement site and catheter size (gauge and length); options included traditional length 24-, 22-, 20-, or 18-gauge IVs or the longer 45-mm length (22-g) or 48-mm length (20-g) catheters.

The same data was obtained for TPIVs placed during the same time period through the EHR. The EHR was examined for a random selection of patients each month with diagnoses that made them likely to receive an IV during their ED visit. These diagnoses included sickle cell disease, diabetes, febrile neonates, and vomiting/dehydration. Providers had the same options for placement site and catheter size of the TPIVs as they did for the USGPIVs.

Analysis

The statistical analyses performed in this study were performed using Stata version 13.1. The survival times of the USGPIVs and the TPIVs placed were calculated based on the time placed and time removed as documented in the EHR. Overall mean survival times of the USGPIVs and the TPIVs were compared with a *t*-test. A Kaplan-Meier survival analysis was performed censoring for IVs removed because they were no longer needed or had incomplete follow-up records. A log-rank test was used to compare the Kaplan-Meier survival curves. The percentage of IVs removed due to a complication was determined, and a chi-squared test was used to compare the complication rates of the USGPIVs versus TPIVs.

Results

Patient Characteristics

From August 2013 to April 2014, a total of 300 USGPIV encounters were recorded. A total of 50 different providers placed the USGPIVs. More than half of the patients (55%) in the USGPIV arm self-reported a history of difficult access, and many of the patients with a successful USGPIV had one (55%) or more than one (38%) previous TPIV attempt during the same encounter. The baseline patient characteristics for the USGPIV group are presented in Table 1, and the outcomes for the USGPIVs are presented in a flowchart (Figure 1).

During the same time period, a total of 552 TPIVs were analyzed. Only 40 (7.2%) self-reported a history of difficult access. The baseline characteristics for the TPIV group are presented in Table 1, and the outcomes for the TPIVs are presented in a flowchart (Figure 2).

IV Survival Times

For the USGPIV encounters, complete follow-up information was available for 160 of the 208 (76.9%) patients that were admitted and had a successful USGPIV placed. Of the 160 USGPIVs with complete follow-up information, the majority (63%) were removed because they were no longer needed. For the TPIV encounters, complete follow-up information was available for 363 of the 428 (85%) patients that were admitted and had a successful TPIV placed. Of the 363 TPIVs with complete follow-up information, the majority (68%) were removed because they were no longer needed. The overall survival time of the USGPIVs placed (mean = 73 hours, SD = 68 hours) was significantly longer than the overall survival time of the TPIVs (mean = 38 hours, SD = 29.4 hours), $t(559) = 8.51$, $P < .0001$.

Table 1. Baseline Patient Characteristics for USGPIVs and TPIVs

Characteristics	USGPIVs	TPIVs
Age		
Mean (SD) —y	12.1 (7.5)	8.32 (7.3)
Range—y	0.05-26.9	0.01-25.7
Age documented—no. (%)	295 (98.3)	552 (100)
Older than 18 y—no. (%)	58 (19.3)	64 (11.6)
IV location—no. (%)		
Forearm	147 (49)	60 (10.9)
Hand	0 (0)	305 (55.3)
Antecubital fossa	36 (12)	49 (8.9)
Upper arm	36 (12)	22 (4.0)
Lower extremity	7 (2.3)	47 (8.5)
Unknown location	46 (15.3)	63 (11.4)
Unsuccessful	28 (9.3)	6 (1.1)
Self-reported history of difficult access—no. (%)		
Yes	165 (55)	40 (7.2)
No	135 (45)	512 (92.8)
Total—no. (%)	**300 (100)**	**552 (100)**

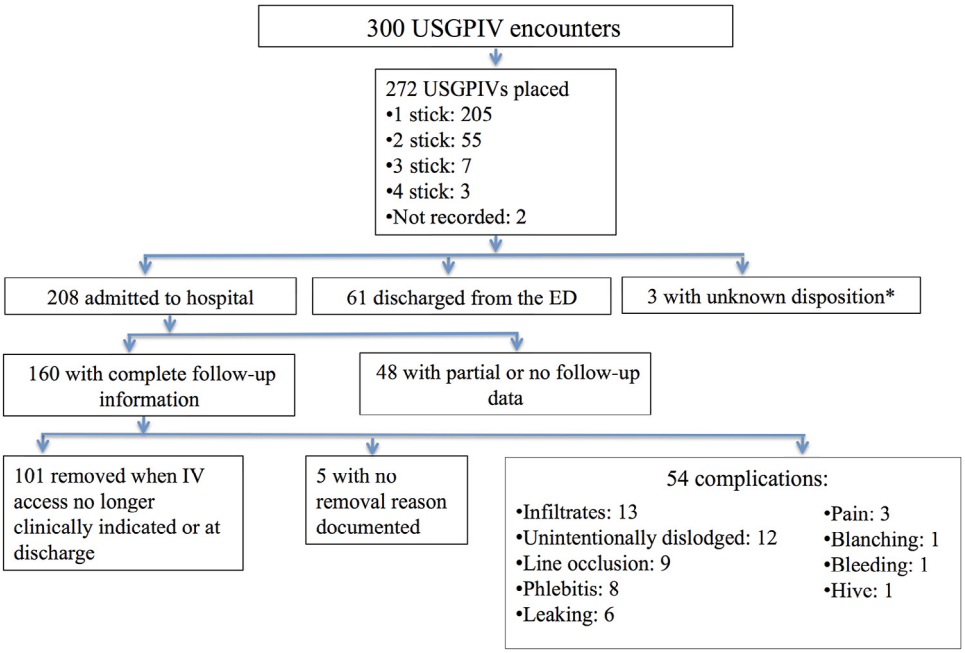

Figure 1. Flowchart of USGPIVs recorded in the study. *Form completed but no patient-identifying information included to enable follow-up.

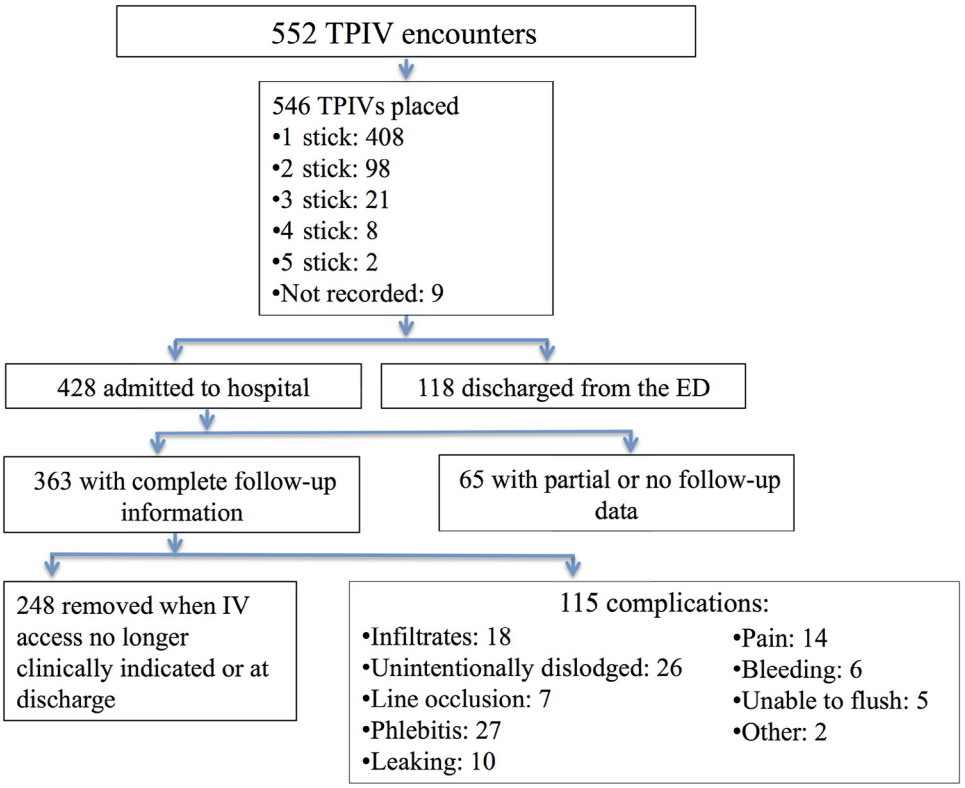

Figure 2. **Flowchart of TPIVs recorded in the study.**

Kaplan-Meier survival curves were generated for both the USGPIVs and TPIVs, censoring for IVs that were no longer needed and those lost to follow-up. The curves for these 2 groups were compared using the log-rank test (Figure 3). The Kaplan-Meier survival time for the USGPIVs was 143 hours (IQR = 68-246), while the Kaplan-Meier survival time for the TPIVs was 89 hours (IQR = 65-106). The log-rank test of equality across the 2 groups showed a significant difference between the survival curves ($P < .001$).

IV Complication Rates

Complication rates for the USGPIVs and TPIVs were calculated both as complications per 1000 catheter days according to INS standards and also as a percentage of the total number

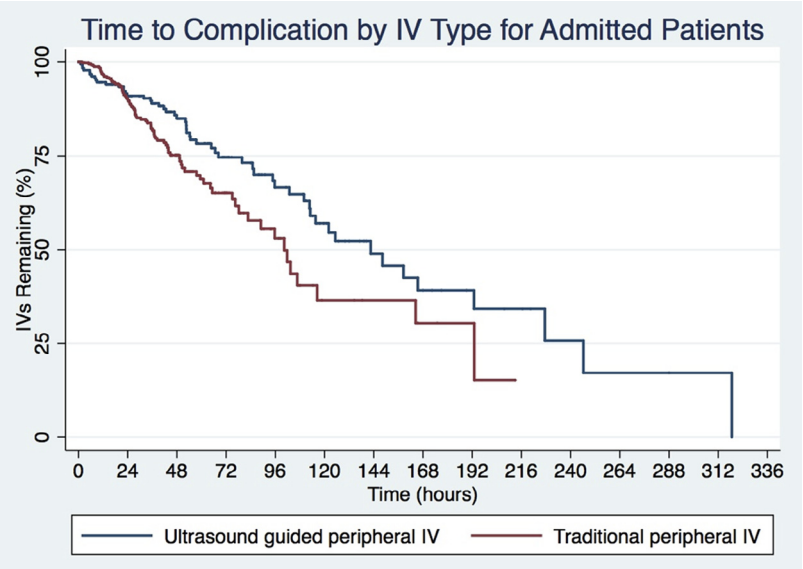

Figure 3. **Kaplan-Meier survival curves for USGPIVs versus TPIVs.**

Table 2. **Removal Reasons for USGPIVs Versus TPIVs**

Removal Reason—No. (%)	USGPIVs	TPIVs
No longer needed	101 (63)	248 (68)
Infiltrates	13 (8.1)	18 (5.0)
Unintentionally dislodged	12 (7.5)	26 (7.2)
Line occlusion	9 (5.6)	7 (1.9)
Phlebitis	8 (5.0)	27 (7.4)
Leakage	6 (3.8)	10 (2.8)
Pain	3 (1.9)	14 (3.9)
Bleeding	1 (6.3)	6 (1.7)
Unable to flush	0 (0)	5 (1.4)
Other	2 (1.3)	2 (0.6)
Removal reason unknown	5 (3.1)	0 (0)
Total	160 (100)	363 (100)

of catheters. For the 160 USGPIVs with complete follow-up data, there were 54 failed USGPIVs with a total number of catheter days equaling 445.4, corresponding to a complication rate of 121.2 complications per 1000 catheter days. As a percentage of the total number of catheters, the complication rate was 34%. Common reasons for IV failure included unintentional dislodgement, infiltration, occlusion, and phlebitis (Table 2). For the 363 successful TPIVs with complete follow-up data, there were 115 complications overall with a total number of catheter days equaling 574.2, corresponding to a complication rate of 200.3 complications per 1000 catheter days. As a percentage of the total number of catheters, the complication rate for the TPIVs was 32%. Common reasons for IV failure mirrored those of the USGPIVs. With either way of reporting the complication rates, there was no significant difference in complication rates between the USGPIVs and TPIVs.

Discussion

While previous studies have reported on USGPIV and TPIV survival times individually, this study is 1 of few to directly compare the longevity of USGPIVs to TPIVs. This study found a mean overall TPIV survival time of 38 hours, which is similar to that reported in studies with overall mean and median survival times ranging from 29-99 hours in both pediatric and adult populations.[9-12] For USGPIVs, previous studies have found overall mean and median survival times ranging from 22-28 hours, which is lower than the overall USGPIV mean survival time of 73 hours found in this study.[9,13,14] The discrepancy between our reported survival times and those that have been previously reported is even more pronounced when using a Kaplan-Meier survival analysis, which censors for IVs removed because they were no longer needed or lost to follow-up. This is a more accurate assessment of peripheral intravenous (PIV) survival times, and future studies should report survival times this way whenever possible.

Our Kaplan-Meier survival time of 89 hours for TPIVs is similar to Kaplan-Meier survival times previously reported in the literature that have ranged from 83-88 hours.[15,16] No prior study to our knowledge has reported Kaplan-Meier survival times for USGPIVs. There are, however, 2 prior studies that have also directly compared the longevity of USGPIVs to TPIVs. One study in adult patients found a significantly longer overall mean survival time of TPIVs (62 hours) compared to USGPIVs (27.91 hours), while the other study was an RCT performed in pediatric patients that found no significant difference between the overall median USGPIV (22.0 hours) and TPIV (23.0 hours) survival time.[9,14] The overall mean survival time and the Kaplan-Meier survival time of the USGPIVs in this study were significantly longer than that of the TPIVs, suggesting that USGPIVs have at least a comparable if not better longevity than TPIVs.

There are a few potential reasons why this study found a longer USGPIV survival time than that of the TPIVs. First, the previous studies only compared overall mean and median survival times rather than Kaplan-Meier survival times. These studies did not censor for IVs removed because they were no longer needed, which can confound survival times by the patient's length of stay rather than purely assessing the durability of their IV access. Second, the longevity of the USGPIVs is likely institution-dependent, and our USGPIVs may have had a longer survival time due to differences in provider training, technique of placement, and experience compared to those at other institutions. The majority of USGPIVs placed at our institution are longer in length than TPIVs, and there are theoretical benefits of a longer catheter to being more securely placed within the vein that may contribute to the longer survival times.

This study found no significant difference in the complication rate between the USGPIVs versus TPIVs. The most common reasons for IV failure were similar to what has been reported in the literature for USGPIVs and TPIVs placed in the pediatric and adult populations, including infiltration, phlebitis, and unintentional dislodgement.[12,17,18] This study did find a lower incidence of IV infiltration for both USGPIVs (8.1%) and TPIVs (5%) compared to infiltration rates of both USGPIVs and TPIVs reported in the literature that range from 14%-28%.[12,15,16] Our phlebitis rates, however, were higher than the phlebitis rates, ranging from 1.5-5% as reported in prior studies.[12,16] As with the survival times, the differences in complication rates can be attributed to institution-dependent differences in placement technique and provider experience or are possibly due to slight differences in the categorization of complications.

Limitations

A limitation of this study is that the USGPIVs analyzed were self-reported via a form completed by the practitioner who was placing the USGPIVs. Because of both the increased time required to fill out the form and possible hesitancy by clinicians to report failures, the number of missed attempts and complications may have been underreported. Nonetheless, all USGPIVs were also monitored as part of a larger quality assurance program

for point-of-care ultrasound exams performed in our ED, and we know that the rates of underreporting were very low.

The patient population that received USGPIVs and TPIVs differed in that those patients that had USGPIVs placed were more likely to have had a history of difficult access reported in the EHR, along with multiple prior unsuccessful needlesticks. In addition, the TPIVs analyzed were chosen based on randomly sampling patients with specific diagnoses that made them likely to receive an IV, whereas the patients with USGPIVs did not necessarily have these specific diagnoses. Having those specific diagnoses may have independently affected the survival times or complication rates of the PIV placed. However, one would expect that since the USGPIV group had a higher proportion of patients with a history of difficult access, the USGPIVs would be more likely to be unsuccessful or have difficult placement that could adversely affect survival and complication rates.

Complete follow-up information was not available for all the patients who received USGPIVs or TPIVs. Patients with incomplete follow-up records were censored based on the time that the IV was last reported as functioning. Therefore, the true survival times for both groups were likely longer than what was reported in our study. Finally, some complications may not have been reported, and the complication rates may have been higher in actuality for either group.

Conclusion

In conclusion, the data in our study suggest that ultrasound is a viable tool for assistance in placing PIVs, particularly in children who have had a history of difficult access. While there are some concerns about the durability of USGPIVs, this study found significantly longer survival times of USGPIVs compared to that of TPIVs—a finding that suggests that USGPIVs last at least as long as TPIVs if not longer. In addition, there was no significant difference in overall complication rates between the USGPIVs and TPIVs, indicating that ultrasound assistance does not introduce any more complications than what would be normally observed. Future studies may include a RCT that compares USGPIVs to TPIVs to better control for differences in the population that receives USGPIVs versus TPIVs.

Disclosures

The authors have no conflicts of interest to disclose.

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